Academic Research Presentation

Inventory Management System for Small Store

Submitted to:

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Table of Contents

1 Background of the Study

2 Problem Statement & Research Questions

3 Objectives

4 Scope of the System

5 Limitations

6 Definition of Terms

7 System Prototype

8 References

Background of the Study

The purpose of this study is to investigate the efficiency of inventory management systems in today's fast-paced retail environment. The shift from traditional pen and paper techniques to digital methods has become a necessity for business survival.

Traditional Challenges:

- Manual ledgers and memory-based tracking work for few items but become hindrances as businesses grow
- Prone to human errors leading to lost sales and capital tied up in unsold products
- Difficulty reconciling physical stock with recorded sales

Digital Transformation:

- Computer-based IMS transforms chaotic processes into organized workflows
- Centralizes product data for real-time stock monitoring
- Enables instant price updates and digital transaction trails
- Provides data for understanding consumer behavior and seasonal trends

Impact on Small Stores:

- Automates inventory counting and report generation
- Allows owners to focus on customer service and strategic growth
- Ensures organized, data-driven decisions with improved accuracy
- Bridges the gap between simple retailing and professionalized management

Statement of the Problem

Small stores often manage their inventory manually, which can lead to problems such as inaccurate stock records, difficulty in tracking products, and time-consuming processes.

Manual inventory systems may also result in errors in monitoring available items, delayed updates, and difficulty in generating reports for sales and stock levels.

> Because of these challenges, store owners may experience problems in maintaining an organized and efficient inventory system.

Research Objective

This study aims to develop an ***Inventory Management System*** *that will help small store owners manage their inventory effectively.*

Research Questions

This study seeks to answer the following questions:

1. How can a computerized system improve the process of managing inventory in a small store?
2. How can the system help store owners track product stock levels efficiently?
3. How can the system reduce errors commonly found in manual inventory management?
4. How can the system generate reports that help store owners monitor sales and inventory?
5. How can the proposed system improve the overall efficiency of inventory management in a small store?

Objectives

GENERAL OBJECTIVE:

The main objective of this study is to develop an **Inventory Management System for a Small Store** that will help improve the monitoring, recording, and management of store products efficiently.

SPECIFIC OBJECTIVES:

1. To create a system that can store and manage product information
2. To create a system that can record and update product information and stock levels efficiently
3. To reduce errors that commonly occur in manual inventory management
4. To generate inventory reports that help in monitoring store operations
5. To improve the overall efficiency and organization of inventory management in a small store
6. To design a user-friendly graphical interface that allows easy interaction with the system

Scope of the System

CORE FUNCTIONALITIES

This research focuses on the development of an Inventory Management System for a Small Store to improve the monitoring and control of store products.

System Capabilities:

- Desktop-based store inventory and sales operations in a single application
- Login screen with name/password inputs and show-password toggle
- Main dashboard with multi-page navigation including Home, Products, Inventory, Sales, and Reports pages

Product Management:

- Product catalog management including add, edit, delete operations with basic validation
- Each product contains Product ID (SKU), Name, Category, Unit price, Quantity, and Reorder level
- Required field validation, numeric validation for price/quantity/reorder level
- Duplicate SKU prevention when adding, and delete confirmation dialog before removal

Scope of the System

ADVANCED FEATURES & REPORTING

Inventory Monitoring:

- Inventory monitoring with stock-status filtering and search capabilities
- Summary cards for Total Products, In Stock, Low Stock, and Out of Stock
- Search by product name or SKU and filter by stock status

Reporting System:

- Operational reporting for products, sales, and inventory
- Three report views: Product Report, Sales Report, and Inventory Report
- Export report data as CSV files with timestamps in filenames

Sales Management:

- Sales transaction recording with stock deduction and sales summaries
- Auto-filled current date, auto-calculated unit price and total, and stock validation before recording

Startup Behavior:

- Demo-ready startup behavior with seeded sample data

Limitations

The advanced features of inventory management, such as ordering suppliers, warehouse management, or connecting to accounting or financial packages, will not be included in the research, nor will it cover advanced data analytics, forecasting, or inventory management for multiple locations. The study will only cover the basic inventory management functions that are required by a small store.

Current Implementation Limitations:

- Authentication is not secure; relies on hardcoded username/password pair (admin / password)
- Runtime data is in-memory only; no database or long-term persistence layer
- Report exports are limited to CSV format only
- Designed for single-user local execution; no multi-user concurrency controls
- Date/time, currency format (Philippine Peso - PHP), and labels are fixed in-app
- Error handling and validation are focused on UI-level checks

Out of Scope:

- Advanced user roles and permissions
- Multi-branch or multi-store inventory synchronization
- Purchase order management and supplier workflows
- Barcode scanning, receipt printing, POS hardware integration
- Cloud/mobile companion applications

Definition of Terms - Part 1

- **Inventory Management System (IMS)**: A computerized system used to track, record, and manage the quantity, movement, and information of products within a store.
- **Product**: An item or merchandise sold in the store containing name, category, price, stock quantity, and identification code.
- **SKU (Stock Keeping Unit)**: A unique code assigned to a product that helps identify and track items in the inventory system.
- **Stock Level**: The current quantity of a product available in the inventory at a given time.
- **Reorder Level**: The minimum quantity of a product that should remain in stock before the store needs to restock.
- **Sale Transaction**: A recorded event where one or more products are sold to a customer.
- **Sale Item**: A specific product included in a sales transaction along with the quantity sold and unit price.
- **Inventory Manager**: A system component responsible for managing product records and updating stock levels.
- **Sales Manager**: A system component responsible for recording sales transactions.

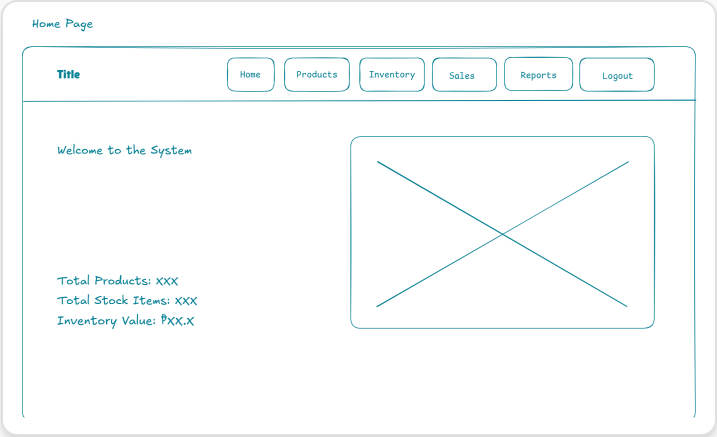
Definition of Terms (Continued)

- **Report Service**: A system feature that generates reports related to inventory and sales.
- **Low Stock**: A condition where quantity falls below the defined reorder level.
- **Out of Stock**: A situation where a product is no longer available (quantity reached zero).
- **User Interface (UI)**: The graphical part of the system that allows users to interact with the program.
- **Graphical User Interface (GUI)**: A type of user interface using graphical elements such as windows, icons, and buttons.
- **Sales Report**: A generated document that summarizes sales transactions over a certain period.
- **Inventory Report**: A report displaying product quantities, stock levels, and inventory status.
- **CSV (Comma-Separated Values)**: A file format used for exporting report data from the system.
- **In-Memory Data**: A data storage approach where information is held in RAM during runtime.
- **Java Swing**: A GUI widget toolkit for Java used to build the desktop interface.
- **Dashboard**: The main overview screen displayed after login providing a quick operational summary.

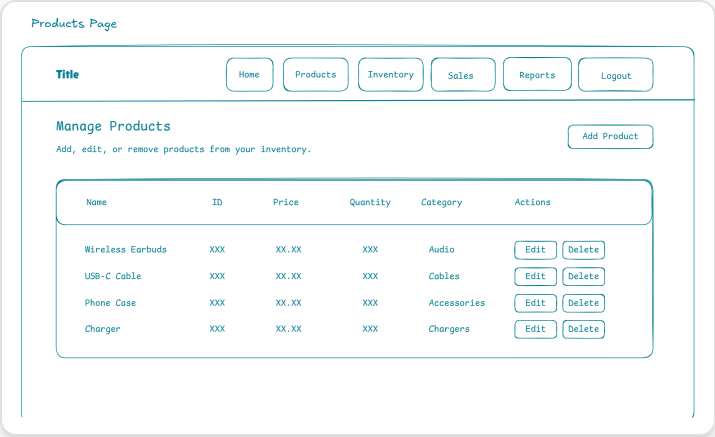
System Prototype

User Interface Screenshots

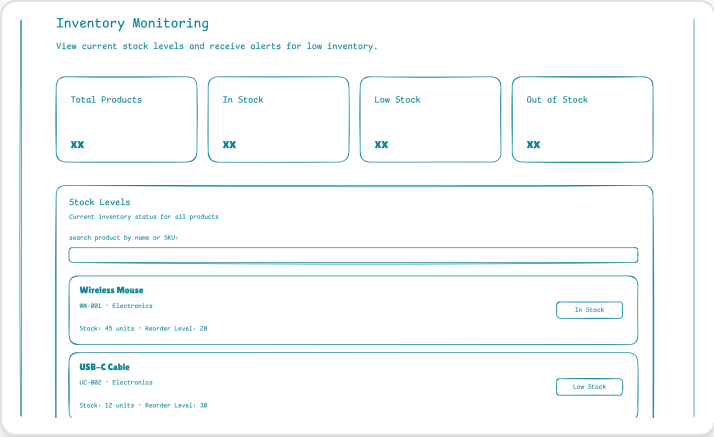
Login Screen



Dashboard Home



Product Management



Add Product Form

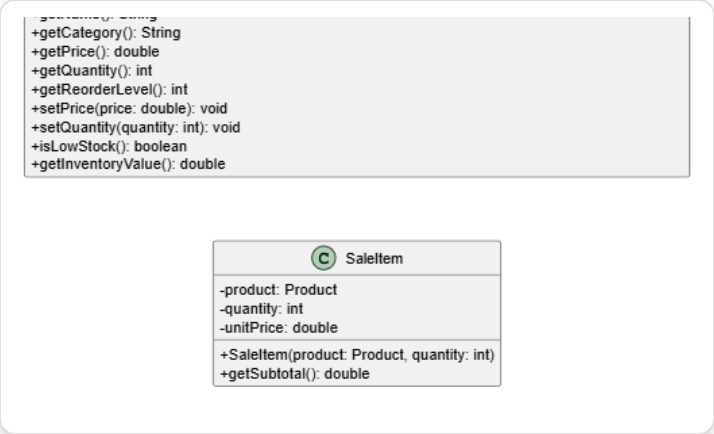
Inventory View

System Prototype (Continued)

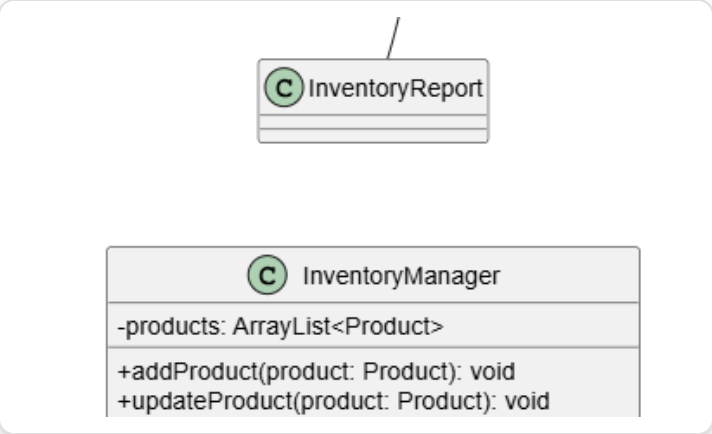
Additional Interface Screenshots

Sales Transaction

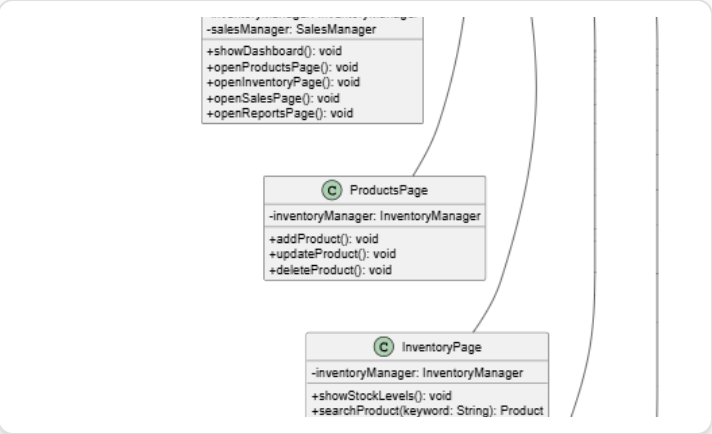
Sales Record



Product Report



Sales Report



Inventory Report

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